The Federal University of Technology, Akure
School of Engineering and Engineering Technology
Department of Information and Communication Engineering
Second Semester Examinations 2023/2024 Session
ICT 305/EEE 311, Electric Circuit Theory; Date: 28th May, 2024: 3 HOURS.

SECTION A

Answer ALL QUESTIONS in this Section

1. The elements which consume energy either by absorbing/storing are called______elements. (A). Active (B) Passive (C) Normal (D) Excited
2. The sources in which the current/voltage does not change with time are called _________.
3. The electrical energy supplied by_____source depends on another source of electrical energy. (A) Passive (B) Independent (C) Active (D) Dependent
4. The path between any two nodes is called: (A) Network (B) Mesh (C) Branch (D) Terminal
5. In an electric circuit when elements are connected in_____ the current will be the same.
6. In an electric circuit a path of infinite resistance is called _________ (A) Open circuit (B) Closed circuit (E) Infinite circuit (D) Short circuit
7. What is the ideal input impedance of an op-amp: (A) Zero (B) 10^{13}kΩ (C) Infinite Ω (D) 50Ω
8. What is the ideal output impedance of an op-amp: (A) Zero (B) 10^{13}kΩ (C) Infinite Ω (D) 50Ω
9. What are the two input terminals of an op-amp: (A) Positive and negative (B) Inverting and non-inverting (C) Ground and power (D) Input and Output
10. What is the primary function of negative feedback in op-amp circuits: (A) Increase distortion (B) Provides stability (C) Increase Bandwidth (D) Reduce input impedance.
11. What is the unit of Capacitance: (A) Volts (B) Farad (C) Ampere (D) Ampere
12. Which parameter determines the energy stored in a capacitor: (A) Voltage (B) Current (C) Resistance (D) Capacitance
13. What happens to the capacitance of a capacitor when the area of the plates is increased? (A) Capacitance decreases (B) Capacitance remains the same (C) Capacitance increases (D) Capacitance becomes zero
14. What is the primary function of a capacitor in an electrical circuit? (A) To amplify signals (B) To store and release energy (C) To regulate voltage (D) To convert AC to DC
15. In terms of the area of the plates and the distance between the plates, state the equation of capacitance.

State whether the following statements are true or false

16. The elements of electric circuits which can deliver energy are called active elements.
17. Inductance and capacitance absorb energy.
18. The electrical energy supplied by an independent source depends on another energy source.
19. In parallel connected elements the voltage will be the same.
20. Capacitors block DC current and allow AC current to pass through
21. The capacitance of a capacitor depends on the voltage across it
22. Capacitors are passive electronic components.
23. Op-amps can only amplify voltage signals
24. Voltage-follower configuration of op-amp provides voltage gain
25. Op-amps can saturate when the output voltage reaches is power supply rails.

SECTION B

Answer ONLY ONE question in this Section

1. (a). Consider the circuit in Figure 1a, sketch out the following from the circuit
   - i). All branches **(3 marks)**
   - ii). All Nodes **(2 marks)**
   - iii). All closed path **(3 marks)**

   (b). i). Find i and V_o in the circuit shown in Figure 1b. All resistances are in Ohms **(6 marks)**
   - ii). In a circuit with a variable resistor, if the resistance is increased, how does it affect the current and voltage in the circuit, assuming the voltage source remains constant. **(4 marks)**

   (c). A series circuit powering three light bulbs is not functioning properly. The voltage across the circuit is 120 volts, and each bulb has a resistance of 10Ω. Use this information to answer the following questions:
   - i). Draw the circuit diagram for this arrangement. **(2 marks)**
   - ii). Determine the total resistance of the circuit. **(2 marks)**
   - iii). Determine the total current flowing through the circuit to trouble shoot the issue. **(3 marks)**

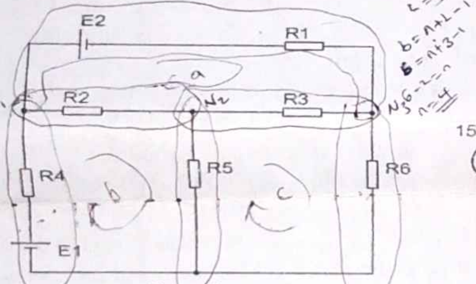

Figure 1a

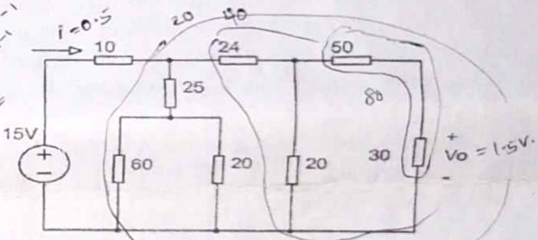

Figure 1b

2. (a). i). Draw out all branches and nodes for the circuit in Figure 2a. **(5 marks)**
   - ii). A circuit contains both series and parallel connections. How does adding a resistor in series affect the total resistance compared to adding the same resistor in parallel. **(4 marks)**

   (b). In the circuit given in Figure 2b, find:
   - i). The total current drawn from the battery. **(3 marks)**
   - ii). The voltage across the 2Ω resistor. **(3 marks)**
   - iii). The current passing through the 5Ω resistor. **(3 marks)**

   (c). In a parallel circuit powering three heaters with resistances of 30Ω each, if the voltage across the circuit is 240 volts, calculate the following:
   - i). Draw the circuit diagram for this arrangement. **(2 marks)**
   - ii). The current through each heater. **(3 marks)**
   - iii). The total power consumed by the circuit. **(2 marks)**

$\frac{1}{R_1}+\frac{1}{R_2} = \frac{R_2+R_1}{R_1R_2} = \frac{1}{R_t}$

$\frac{30+30}{30+30+30} = \frac{64}{96} \times 24\ 8 = 16A.$

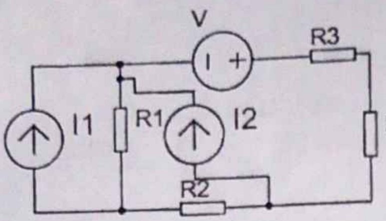

Figure 2a

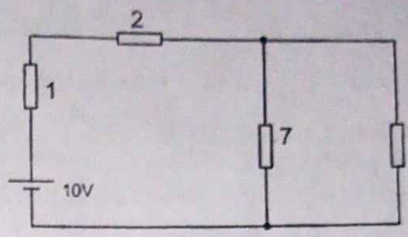

Figure 2b

b=4+n-1
5=3+n-1
5-2=n

SECTION C

Answer ONLY ONE question in this Section

3. **(a).** Consider the circuit given in Figure 3a, use mesh analysis to find the value of E such that $I_2 = 0$. **(9 marks)**

   (b). Using Super mesh analysis, find the power dissipation across the 4Ω Resistor shown in the circuit in Figure 3b. **(10 marks)**

   (c). With diagram(s), describe the concept of op-amp voltage follower and give one useful application in circuit design. **(6 marks)**

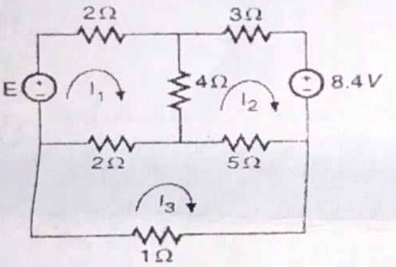

Figure 3a

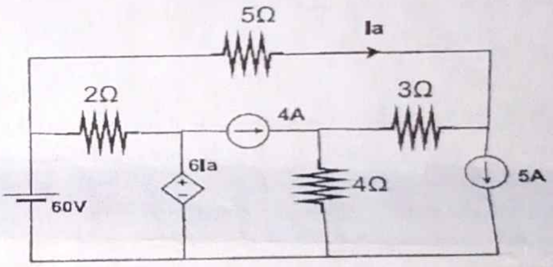

Figure 3b

4. **(a).** Determine the power dissipation in the 4Ω Resistor of the circuit shown in Figure 4a. Use Mesh Analysis and KCL. **(8 marks)**

   (b). In the circuit shown in Figure 4b, find the current supplied by the voltage source and the voltage across the current source using super mesh analysis. **(10 marks)**

   (c). With diagram(s), describe how an ideal op-amps are configured to operate as an inverting amplifier. **(7 marks)**

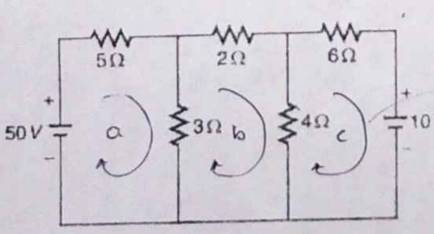

Figure 4a

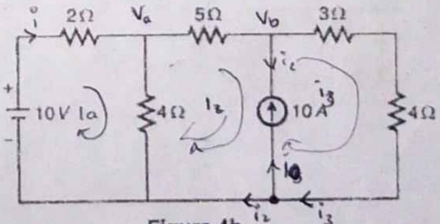

Figure 4b

$-50+5i_a+3(i_a-i_b)$

$\boxed{8i_a-3i_b=50}$ — ①

$4(i_c-i_b)+4i_c+10=0$

$4i_c-4i_b+8i_c=-10$

$-4i_b+8i_c=-10$

$\boxed{-2i_b+5i_c=-5}$

$3(i_b-i_a)+2i_b+4(i_b-i_c)$ 3

$3i_b-3i_a+2i_b+4i_b-4i_c=0$

$\boxed{-3i_a+9i_b-4i_c=0}$

$-10+2i_a+4(i_a-i_b)=0$

$2i_a+4i_a-4i_b=10$

$6i_a-4i_b=10$

$3i_a-2i_b=5$ — ①

$4(i_b-i_a)+5i_b+3i_c+4i_c$

$4i_b-4i_a+5i_b+3i_c+4i_c$

$-4i_a+9i_b+7i_c=0$

$i_3=10+i_2$

$i_2=10+i_3$

$-4.75=10+5.25$

$-4.75-5.25=10$

$-10=10$

SECTION D

Answer ONLY ONE question in this Section

5. **(a).** Consider the circuit given in Figure 5a, Find the voltage across the 40Ω Resistor and the power supplied by the 5A source. Use Super node Analysis. **(10 marks)**

   (b). Calculate the gain v_o/v_i when the switch is in:
   i). Position 1 **ii).** Position 2, **iii).** Position 3 **(6 marks)**

   (c). Determine the current through a 200µF capacitor whose voltage is shown in Figure 5c **(9 marks)**

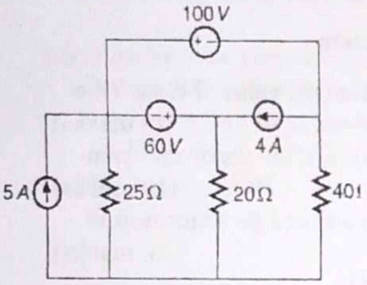

Figure 5a

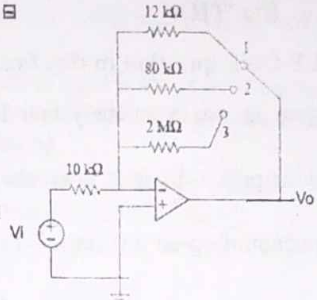

Figure 5b

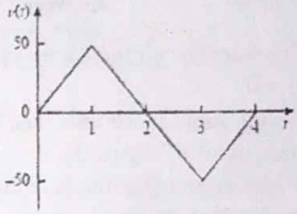

Figure 5c

$\frac{di}{dt} = C\frac{dv}{dt}$

$i\,dt = C\,dv$

$\frac{1}{C}\int i\,dt = \int dv$

$\frac{1}{C}\int i\,dt =$

6. **(a).** Using Super node analysis, calculate I_o in the circuit given in Figure 6a **(10 marks)**

   (b). For the op-amp circuit given in Figure 6b, determine the value of V_2 in order to make V_o = -16.5 V. **(7 marks)**

   (c). i). Find the equivalent capacitance seen at the terminal if the circuit given in Figure 6c. **(4 marks)**

   ii). Determine the voltage across a 2µF capacitor if the current through it is

$$i(t) = 6e^{-3000t}\ mA$$

   Assume that the initial capacitor voltage is zero **(4 marks)**

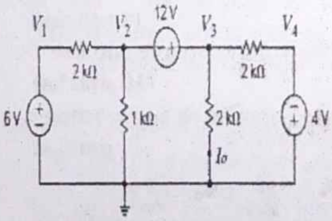

Figure 6a

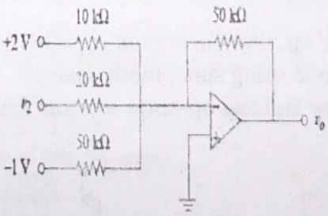

Figure 6b

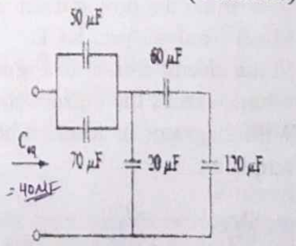

Figure 6c